

# The Agony and the Ecstasy

Constructing a “Crash-Filtered” Equity Index using Machine Learning

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# The Agony and Ecstasy

## Passive indices are exposed to both extremes

Long-run index returns are driven by a few high-performing stocks, while a considerable share of constituents suffer catastrophic declines.

### Ecstasy

- Extreme winners carry a large share of long-run wealth creation.
- Winners can be volatile, expensive, or distressed.
- Thus, even a small amount of disregarded winners can hurt performance.

### Agony

- Some constituents experience deep and persistent drawdowns without recovery.
- Passive strategies may hold imploding firms for too long.

## Solution

The challenge is not to eliminate highly volatile stocks. It is to distinguish recoverable volatility from persistent implosion. The empirical question is whether catastrophically declining firms are forecastable using ideas from credit-risk modelling, but with market-based distress signals.

# Research Question

## Main research question

To what extent does a “Crash-Filtered” equity index, constructed via probabilistic implosion modeling, generate superior risk-adjusted returns compared to the market benchmark and traditional minimum-volatility strategies?

The probabilistic implosion model follows the Catastrophic Stock Implosion approach proposed by Tewari et al. (2024). Instead of relying only on formal bankruptcy declarations, CSI uses market-based price-path signals.

## Subquestions:

- How does the integration of Autoencoders for feature engineering impact the Average Precision of Ensemble models compared to those trained solely on raw financial data?
- To what extent do Ensemble methods reduce the False Positive Rate (classifying recoverable volatility as CSI) compared to traditional volatility-based exclusion strategies, while maintaining Recall?
- Which features are most important for distinguishing between “Zombie” firms (CSI) and non-zombie firms?

## Empirical design

The final panel keeps only observable CRSP firm-years: each row has the required monthly return and market-cap scaffold. Temporary and permanent CSI use the same 188,460-row scaffold; unresolved  $y = \text{NA}$  rows are retained in the panel but excluded from supervised model metrics.

### Data sources

#### CRSP

monthly returns, market cap, delistings, wealth paths

#### Compustat

annual accounting fundamentals

#### FRED

macro variables

### Universe

#### Minimum size

market cap  $\geq$  USD 100M

#### Observable scaffold

188,460 annual firm-year rows in both tracks

#### Index buckets

Total, Large, Mid, Small Cap

### Sample period

#### Coverage

Jan 1993 – Dec 2024 (32y, monthly frequency)

#### Supervised labels

train/test labelled; OOS may retain unresolved labels

*metrics use labelled rows only*

## Temporal split



# Methodology I: Response Variable Classification

Following Tewari et al. (2024), a stock is classified as a CSI candidate if it satisfies all three path criteria:

- **Initial Crash ( $C$ ):** more than a  $C$  drawdown from the trailing peak marks the beginning of the “zombie” period.  $C \leq -80\%$
- **Non-Recovery ( $M$ ):** maximum cumulative return remains below the recovery ceiling  $M$  during the zombie period.  $M \leq -20\%$
- **Zombie Period ( $T$ ):** duration of the confirmation window during which non-recovery must hold.  $T = 18 \text{ months}$

The prediction task is annual. A trigger in calendar year  $t + 1$  gives the firm-year observation at  $t$  a positive label when the track-specific confirmation rule is met:

$$y_{i,t} = \begin{cases} 1, & \text{if stock } i \text{ triggers a qualifying CSI event within } [t, t + h], \\ 0, & \text{otherwise.} \end{cases}$$

## Two final response tracks

**Temporary CSI** enables the accepted terminal-failure route and keeps censored triggers plus unavailable 2025 event-year labels as  $y = \text{NA}$ . **Permanent CSI** uses event-window-only unresolved PCL windows; broad no-trigger calendar censoring is not used.

# Applying the classification: Temporary CSI

## Temporary CSI

Terminal-failure route enabled. The temporary-CSI label has 8,517 positive annual cells in the full observable scaffold. The 8,674 OOS  $y = NA$  rows are unresolved labels: 3,342 censored trigger labels plus 5,332 observable 2024 rows that would require unavailable 2025 event-year labels.

| Split | Obs. rows | $y = 0$ | $y = 1$      | $y = NA$ | Labelled | Prev. |
|-------|-----------|---------|--------------|----------|----------|-------|
| Full  | 188,460   | 171,269 | <b>8,517</b> | 8,674    | 179,786  | 4.74% |
| Train | 143,173   | 136,630 | 6,543        | 0        | 143,173  | 4.57% |
| Test  | 18,111    | 17,307  | 804          | 0        | 18,111   | 4.44% |
| OOS   | 27,176    | 17,332  | 1,170        | 8,674    | 18,502   | 6.32% |

## Motivation for robustness

Prevalence is computed on labelled cells only:  $y = 1/(y = 0 + y = 1)$ . The retained  $y = NA$  rows are observable firm-years with unresolved labels, not dropped rows; they are excluded from supervised training and label-based evaluation.

# Applying the classification: Permanent CSI

## Permanent CSI

Permanent CSI uses the same 188,460 observable firm-year scaffold. Unresolved labels are limited to event-window PCL cases; broad no-trigger calendar censoring is explicitly not used.

| Split | Obs. rows | $y = 0$ | $y = 1$      | $y = NA$ | Labelled | Prev. |
|-------|-----------|---------|--------------|----------|----------|-------|
| Full  | 188,460   | 181,368 | <b>6,258</b> | 834      | 187,626  | 3.34% |
| Train | 143,173   | 137,794 | 5,379        | 0        | 143,173  | 3.76% |
| Test  | 18,111    | 17,511  | 542          | 58       | 18,053   | 3.00% |
| OOS   | 27,176    | 26,063  | 337          | 776      | 26,400   | 1.28% |

## Metric denominator

For both tracks, labelled rows are  $y = 0 + y = 1$  and prevalence is  $y = 1 / (y = 0 + y = 1)$ . Unresolved rows remain in the cleaned panel but are excluded from supervised training and evaluation metrics.

# Robustness I: Temporary CSI Sensitivity Grid

## Completed local sensitivity evidence

The temporary-CSI C/M/T sensitivity grid is available locally. All 27 run IDs are represented; 24 have complete/reused compact results and 3 are explicitly marked `blocked_partial`. Derived presentation summaries remain local-only under `03_Data_Output`.

### Grid design

- Crash threshold  $C$ :  $-60\%$ ,  $-80\%$ ,  $-90\%$ .
- Recovery threshold  $M$ :  $0\%$ ,  $-20\%$ ,  $-30\%$ .
- Confirmation window  $T$ : 12, 18, 28 months.
- Track: temporary CSI only; raw model sensitivity.

### Local status

| Status                       | Runs |
|------------------------------|------|
| Represented in registry      | 27   |
| Complete or safely reused    | 24   |
| <code>blocked_partial</code> | 3    |
| Checksum failures            | 0    |

## Blocked partial runs

`C080_M000_T012`, `C080_M000_T018`, and `C060_M020_T028` are listed as `blocked_partial`; they are not treated as completed evidence.

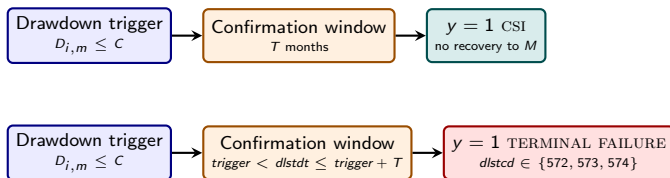
*Remark:* Sources: `local_presentation_ready/sensitivity.cmt_model.summary.csv`; `temporary_blocked.config.disclosure.csv`; AE-SENS-PRES-003 evidence.



# Methodology II: “Temporary-CSI”

## Revision

Original CSI rule unchanged. Temporary CSI uses the accepted CSI\_POSITIVE\_EVENT\_STATUSES; a terminal-failure route is added only after a valid drawdown trigger. Censored triggers and unavailable 2025 event-year labels are retained as  $y = \text{NA}$ .



| Scenario                      | Bankrupting | Detected   | Missed | Detection     |
|-------------------------------|-------------|------------|--------|---------------|
| Before: CSI only              | 629         | 151        | 478    | 24.0%         |
| After: CSI + terminal failure | 629         | <b>545</b> | 84     | <b>86.65%</b> |

The 84 remaining misses are treated as methodological exclusions under the retained CSI-trigger rule, not as known pipeline errors.

# Robustness II: Sensitivity Results and Limits

Sensitivity winners point to different objectives, not one universal setting

| Objective | Config         | CV<br>AP     | CV<br>AUC | CV<br>FPR3 | Test<br>AP   | Test<br>AUC | Test<br>FPR3 |
|-----------|----------------|--------------|-----------|------------|--------------|-------------|--------------|
| Composite | C090_M000_T012 | 0.442        | 0.916     | 0.370      | 0.438        | 0.917       | 0.347        |
| AP winner | C060_M000_T012 | <b>0.473</b> | 0.825     | 0.171      | <b>0.476</b> | 0.859       | 0.173        |
| 11C TM    | C090_M020_T018 | 0.198        | 0.897     | 0.279      | 0.209        | 0.906       | 0.271        |
| Baseline  | C080_M020_T018 | 0.203        | 0.862     | 0.230      | 0.199        | 0.877       | 0.200        |

## Interpretation

The baseline remains defensible as a continuity setting, but it is not the top-ranked sensitivity configuration. CV/test metrics show that the AP winner is not the index-alpha winner, so predictive ranking and portfolio performance are related but distinct objectives.

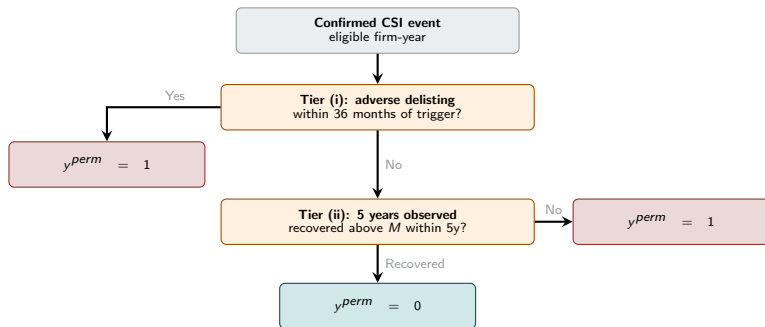
Use sensitivity as robustness evidence, not as a replacement for the final index suite. OOS index-alpha comparisons are shown later in the index and robustness sections. *Remark:* Sources: local

`sensitivity.cmt.model.summary.csv`; `temporary.presentation.key.findings.csv`. FPR3 columns are R@FPR3.

# Methodology III: “Permanent-CSI”

## Dealing with recovering firms

Permanent-CSI starts from a confirmed CSI event and separates terminal failure, durable non-recovery, resolved recovery, and unresolved PCL windows. Its  $y = \text{NA}$  policy is event-window-only; broad no-trigger calendar censoring is not used.



## Contrast with temporary CSI

Temporary CSI applies a finite lockout after a flag. Permanent-CSI marks terminal failure or durable non-recovery as  $y^{perm} = 1$ , resolved recovery as  $y^{perm} = 0$ , and unresolved PCL windows as  $y = \text{NA}$ . The final full panel has 6,258 positives across 187,626 labelled firm-years (3.34%).

# Dataset: Descriptive Statistics

## CSI positives are rare and economically distinct from non-positives

Both label definitions isolate a small distressed tail. The key contrast is not only prevalence: positive firms are much smaller and show worse realized returns and profitability.

| Sample | CSI   | y | Obs.         | Share  | Firms  | Median<br>mcap | Ann.<br>ret.  | ROA           |
|--------|-------|---|--------------|--------|--------|----------------|---------------|---------------|
| Full   | Temp. | 0 | 171,269      | 95.26% | 19,073 | 254.9          | 4.9%          | 1.6%          |
| Full   | Temp. | 1 | <b>8,517</b> | 4.74%  | 5,603  | <b>74.9</b>    | <b>-35.0%</b> | <b>-18.8%</b> |
| Full   | Perm. | 0 | 181,368      | 96.66% | 19,565 | 252.6          | 4.0%          | 1.5%          |
| Full   | Perm. | 1 | 6,258        | 3.34%  | 4,696  | <b>65.7</b>    | <b>-37.5%</b> | <b>-20.0%</b> |

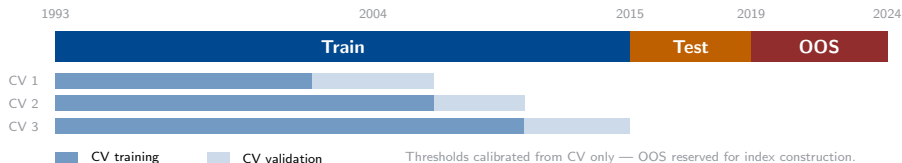
## Read

The full temporary-CSI classification contains **8,517** positive annual cells. Positive labels identify much smaller firms with sharply negative returns and profitability.

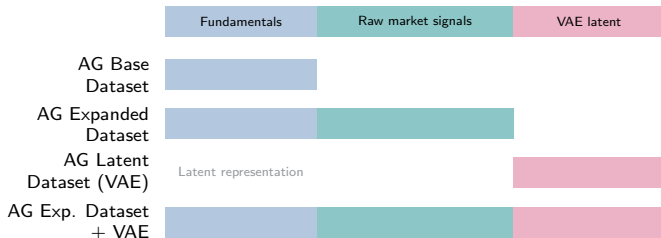
*Remark:* Obs. are annual label cells; Firms are unique firms. Market cap is USD millions.

# Modelling I: Setup and Feature Engineering

## Validation design



## Final feature-set keys



# Modelling II: Results Temporary CSI

## Main read

Temporary CSI remains highly rankable, but AP and fixed-FPR recall are the decision metrics because positives are rare. **AG Exp. Dataset + VAE** is strongest on train/CV and OOS AP/FPR recall; **AG Expanded Dataset** remains the strongest 2016–2019 test benchmark and has marginally higher OOS AUC.

| Model                   | CV-AP         | CV-AUC        | CV-FPR3       | Test-AP       | Test-AUC      | Test-FPR3     |
|-------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| AG Expanded Dataset     | 0.2046        | 0.8635        | 0.2344        | <b>0.1985</b> | <b>0.8766</b> | <b>0.2002</b> |
| AG Base Dataset         | 0.1923        | 0.8446        | 0.2206        | 0.1656        | 0.8538        | 0.1517        |
| AG Latent Dataset (VAE) | 0.1659        | 0.8454        | 0.1706        | 0.1501        | 0.8438        | 0.1294        |
| AG Exp. Dataset + VAE   | <b>0.2114</b> | <b>0.8667</b> | <b>0.2478</b> | 0.1864        | 0.8741        | 0.1779        |

## Interpretation

VAE latent information helps most when added to the expanded feature set and evaluated OOS. The expanded-dataset benchmark remains competitive, so model slides alone do not establish index-level superiority; the index section evaluates economic performance.

*Remark:* CV-FPR3 and Test-FPR3 are R@FPR3. OOS, R@FPR1, R@FPR5, Brier, and full split tables are in Appendix A10.

# Modelling III: Results Permanent-CSI

## Main read

Permanent CSI is harder: positives are rarer and OOS prevalence is much lower. **AG Exp. Dataset + VAE** is the best non-raw CV/test challenger, while **AG Latent Dataset (VAE)** is the strongest OOS robustness signal.

| Model                   | CV-AP         | CV-AUC        | CV-FPR3       | Test-AP       | Test-AUC      | Test-FPR3     |
|-------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| AG Expanded Dataset     | 0.1854        | <b>0.8757</b> | <b>0.2607</b> | <b>0.1416</b> | 0.8810        | 0.1808        |
| AG Base Dataset         | 0.1785        | 0.8677        | 0.2379        | 0.1289        | 0.8627        | 0.1697        |
| AG Latent Dataset (VAE) | 0.1426        | 0.8460        | 0.1960        | 0.1115        | 0.8496        | 0.1513        |
| AG Exp. Dataset + VAE   | <b>0.1883</b> | 0.8719        | 0.2578        | 0.1415        | <b>0.8838</b> | <b>0.2011</b> |

## Interpretation

AG Expanded Dataset remains a strong reporting baseline. For permanent CSI, AG Exp. Dataset + VAE is the primary non-raw challenger, while AG Latent Dataset (VAE) is retained for OOS robustness rather than treated as a clean replacement.

*Remark:* CV-FPR3 and Test-FPR3 are R@FPR3. OOS, R@FPR1, R@FPR5, Brier, and full split tables are in Appendix A11.

# Modelling IV: What VAE Features Add

## Feature-set interpretation

The VAE does not replace expanded-dataset predictive information. It is useful as an augmentation or robustness signal: AG Exp. Dataset + VAE improves temporary OOS AP/FPR recall, while AG Latent Dataset (VAE) is the strongest permanent OOS model.

### Temporary CSI

- AG Exp. Dataset + VAE leads OOS AP: 0.3152.
- OOS R@FPR1/3/5 improves to 0.1051 / 0.2632 / 0.4128.
- AG Expanded Dataset still has the slightly higher OOS AUC: 0.8961 vs 0.8950.

### Permanent CSI

- AG Exp. Dataset + VAE leads test AUC and FPR3/5 recall.
- AG Latent Dataset (VAE) leads OOS AP/AUC/Brier/FPR recall.
- AG Base Dataset remains a useful fundamentals-only benchmark, not the preferred model.

## Model-family note

Weighted AutoGluon ensembles top the leaderboard across tracks and feature sets. Tree families supply most of the useful base-model signal; neural nets are considered but do not dominate the final ranks.

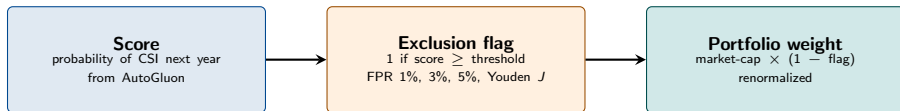


# Index Construction I: From Score to Portfolio Weight

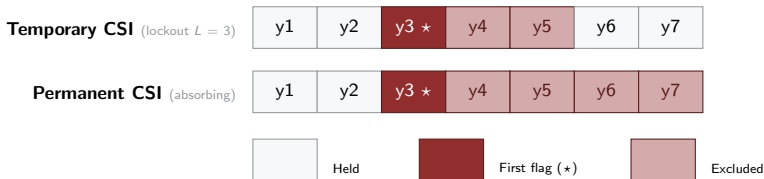
## One score rule, two exclusion regimes

The model produces a probability of CSI next year. A threshold turns that probability into a binary exclusion flag. The flag is then applied to market-cap weights. Two regimes differ only in how long the flag persists after it first fires.

### From model score to portfolio weight



### How the flag evolves over time — a firm crosses the threshold in year 3



# Index Construction II: Four Benchmark Universes

## Same exclusion rule, four universes

The revised 11C track applies the score-to-weight pipeline independently inside each CRSP-like universe, producing four screened benchmarks alongside their unfiltered counterparts.

### Total Market

full universe  
≥ USD 100M

### Large Cap

CRSP-like  
top tier

### Mid Cap

CRSP-like  
middle tier

### Small Cap

CRSP-like  
bottom tier

## Benchmark construction

- Universe formed from prior-December market capitalization.
- Minimum market cap: USD 100 million.
- Quarterly rebalancing: March, June, September, December.
- Market-cap weights; cost overlays use 0, 5, 10, and 20 bps.

## Signal discipline

- Scores are annual for `raw`, `fund`, `latent_raw`, and `raw_plus_latent`.
- Holding-year signal uses information available before the holding year.
- Thresholds calibrated from CV predictions only: Youden, FPR1, FPR3, FPR5.

# Index Results: Temporary CSI at 0 bps

OOS benchmark and best strategy by universe; no transaction-cost overlay

| Universe | Strategy                          | Geo ret. | Ann. SD | Sharpe Ratio | Max DD  | ES 2.5% | Delta pp |
|----------|-----------------------------------|----------|---------|--------------|---------|---------|----------|
| Total    | Benchmark                         | 13.29%   | 19.03%  | 0.598        | -25.03% | -11.62% | 0.00pp   |
| Total    | AG Base Dataset; Youden/3yr       | 13.72%   | 18.28%  | 0.636        | -23.78% | -11.22% | +0.43pp  |
| Large    | Benchmark                         | 14.28%   | 18.49%  | 0.657        | -24.79% | -10.84% | 0.00pp   |
| Large    | AG Base Dataset; Youden/3yr       | 14.43%   | 17.98%  | 0.678        | -23.89% | -10.65% | +0.15pp  |
| Mid      | Benchmark                         | 9.65%    | 20.72%  | 0.406        | -24.58% | -13.83% | 0.00pp   |
| Mid      | AG Base Dataset; FPR5/5yr         | 10.16%   | 20.70%  | 0.429        | -24.46% | -13.85% | +0.51pp  |
| Small    | Benchmark                         | 7.65%    | 23.80%  | 0.305        | -29.05% | -16.12% | 0.00pp   |
| Small    | AG Exp. Dataset + VAE; Youden/3yr | 8.29%    | 22.65%  | 0.336        | -29.33% | -15.77% | +0.64pp  |

## Temporary CSI conclusion

Without transaction costs, AG Base Dataset wins Total, Large, and Mid Cap. AG Exp. Dataset + VAE wins Small Cap. Delta pp is strategy net geometric return minus the matching benchmark row.

Remark: Source: `best.by.track.index.cost.csv`; Ann. SD and ES 2.5% from `model-specific.index.performance.gross.and.net.by.tc.csv`.

# Index Results: Temporary CSI at 10 bps

OOS benchmark and best strategy by universe; 10 bps cost rows

| Universe | Strategy                          | Geo ret. | Ann. SD | Sharpe Ratio | Max DD  | ES 2.5% | Delta pp |
|----------|-----------------------------------|----------|---------|--------------|---------|---------|----------|
| Total    | Benchmark                         | 13.29%   | 19.03%  | 0.598        | -25.03% | -11.62% | 0.00pp   |
| Total    | AG Base Dataset; Youden/3yr       | 13.71%   | 18.28%  | 0.635        | -23.79% | -11.22% | +0.42pp  |
| Large    | Benchmark                         | 14.28%   | 18.49%  | 0.657        | -24.79% | -10.84% | 0.00pp   |
| Large    | AG Base Dataset; Youden/3yr       | 14.42%   | 17.98%  | 0.677        | -23.90% | -10.65% | +0.14pp  |
| Mid      | Benchmark                         | 9.65%    | 20.72%  | 0.406        | -24.58% | -13.83% | 0.00pp   |
| Mid      | AG Base Dataset; FPR5/5yr         | 10.05%   | 20.70%  | 0.424        | -24.46% | -13.85% | +0.39pp  |
| Small    | Benchmark                         | 7.65%    | 23.80%  | 0.305        | -29.05% | -16.12% | 0.00pp   |
| Small    | AG Exp. Dataset + VAE; Youden/3yr | 8.20%    | 22.64%  | 0.332        | -29.33% | -15.77% | +0.55pp  |

## Temporary CSI transaction-cost read

The 10 bps overlay leaves the temporary CSI winners unchanged. Costs apply to gross buy+sell turnover; benchmark rows stay at the unfiltered market-cap reference with no strategy cost overlay.

*Remark:* Strategy rows use source-provided 10 bps net performance; expected annual drag is approximately annualized gross turnover  $\times$  0.001. Source: `best_by_track_index_cost.csv`.

# Temporary CSI: Error-Cost Decomposition

## OOS alpha decomposition for the selected temporary-CSI best strategies

| Universe | Best rule           | FP cost | FN cost | TP gain | TN gain | Net     |
|----------|---------------------|---------|---------|---------|---------|---------|
| Total    | AG Base, Youden 3yr | -2.03pp | -0.00pp | +0.08pp | +1.79pp | +0.43pp |
| Large    | AG Base, Youden 3yr | -1.50pp | +0.00pp | -0.00pp | +1.24pp | +0.15pp |
| Mid      | AG Base, FPR5 5yr   | +0.05pp | +0.00pp | -0.00pp | +0.27pp | +0.51pp |
| Small    | AG +VAE, Youden 3yr | -4.00pp | -0.02pp | +0.42pp | +3.97pp | +0.64pp |

## Why temporary CSI outperforms

The strongest gains come from preserving benchmark exposure to correctly retained firms, while false-positive exclusions are the main offset. Small Cap adds visible true-positive gains, but the net result still depends on avoiding too much opportunity cost.

FP and FN are opportunity-cost terms; TP and TN are gains versus the matched benchmark decomposition. Net matches the selected OOS strategy alpha before cost overlays. *Remark:* Source: model-specific

`error_cost_decomposition_by_crsp_universe.csv`; matched to `best_by_track_index_cost.csv` by track, universe, model, threshold, lockout, strategy id, and OOS period.

# Index Results: Permanent CSI at 0 bps

OOS benchmark and best strategy by universe; permanent-removal rule

| Universe | Strategy                               | Geo ret. | Ann. SD | Sharpe Ratio | Max DD  | ES 2.5% | Delta pp |
|----------|----------------------------------------|----------|---------|--------------|---------|---------|----------|
| Total    | Benchmark                              | 13.29%   | 19.03%  | 0.598        | -25.03% | -11.62% | 0.00pp   |
| Total    | AG Exp. Dataset + VAE;<br>FPR5/perm.   | 13.56%   | 18.82%  | 0.615        | -24.72% | -11.45% | +0.27pp  |
| Large    | Benchmark                              | 14.28%   | 18.49%  | 0.657        | -24.79% | -10.84% | 0.00pp   |
| Large    | AG Exp. Dataset + VAE;<br>FPR5/perm.   | 14.51%   | 18.38%  | 0.671        | -24.51% | -10.74% | +0.23pp  |
| Mid      | Benchmark                              | 9.65%    | 20.72%  | 0.406        | -24.58% | -13.83% | 0.00pp   |
| Mid      | AG Base Dataset; FPR5/perm.            | 10.39%   | 20.58%  | 0.441        | -24.52% | -13.90% | +0.74pp  |
| Small    | Benchmark                              | 7.65%    | 23.80%  | 0.305        | -29.05% | -16.12% | 0.00pp   |
| Small    | AG Latent Dataset (VAE);<br>FPR3/perm. | 7.98%    | 23.78%  | 0.318        | -29.26% | -16.35% | +0.32pp  |

## Permanent CSI conclusion

Without transaction costs, AG Exp. Dataset + VAE wins Total and Large Cap; AG Base Dataset wins Mid Cap; AG Latent Dataset (VAE) wins Small Cap.

*Remark:* Benchmark rows are model-key bench, strategy bench\_mw, period oos. Source: model-specific index\_performance\_gross\_and\_net\_by\_tc.csv.

# Index Results: Permanent CSI at 10 bps

OOS benchmark and best strategy by universe; 10 bps cost rows

| Universe | Strategy                               | Geo ret. | Ann. SD | Sharpe Ratio | Max DD  | ES 2.5% | Delta pp |
|----------|----------------------------------------|----------|---------|--------------|---------|---------|----------|
| Total    | Benchmark                              | 13.29%   | 19.03%  | 0.598        | -25.03% | -11.62% | 0.00pp   |
| Total    | AG Exp. Dataset + VAE;<br>FPR5/perm.   | 13.55%   | 18.82%  | 0.614        | -24.72% | -11.45% | +0.26pp  |
| Large    | Benchmark                              | 14.28%   | 18.49%  | 0.657        | -24.79% | -10.84% | 0.00pp   |
| Large    | AG Exp. Dataset + VAE;<br>FPR5/perm.   | 14.49%   | 18.38%  | 0.670        | -24.51% | -10.74% | +0.22pp  |
| Mid      | Benchmark                              | 9.65%    | 20.72%  | 0.406        | -24.58% | -13.83% | 0.00pp   |
| Mid      | AG Base Dataset; FPR5/perm.            | 10.28%   | 20.58%  | 0.436        | -24.52% | -13.90% | +0.62pp  |
| Small    | Benchmark                              | 7.65%    | 23.80%  | 0.305        | -29.05% | -16.12% | 0.00pp   |
| Small    | AG Latent Dataset (VAE);<br>FPR3/perm. | 7.90%    | 23.78%  | 0.315        | -29.26% | -16.35% | +0.24pp  |

## Permanent CSI transaction-cost read

The 10 bps overlay leaves the permanent CSI winners unchanged. Costs apply to gross buy+sell turnover; benchmark rows stay at the unfiltered market-cap reference with no strategy cost overlay.

*Remark:* Strategy rows use source-provided 10 bps net performance; expected annual drag is approximately annualized gross turnover  $\times$  0.001. Delta pp compares against the unchanged benchmark row.

# Permanent CSI: Error-Cost Decomposition

OOS alpha decomposition for the selected permanent-CSI best strategies

| Universe | Best rule             | FP cost | FN cost | TP gain | TN gain | Net     |
|----------|-----------------------|---------|---------|---------|---------|---------|
| Total    | AG +VAE, FPR5 perm.   | -0.49pp | -0.00pp | +0.01pp | +0.52pp | +0.27pp |
| Large    | AG +VAE, FPR5 perm.   | -0.26pp | +0.00pp | +0.00pp | +0.30pp | +0.23pp |
| Mid      | AG Base, FPR5 perm.   | -0.55pp | +0.00pp | +0.00pp | +0.90pp | +0.74pp |
| Small    | AG Latent, FPR3 perm. | -0.74pp | -0.00pp | +0.04pp | +1.03pp | +0.32pp |

## Why permanent CSI outperforms

The permanent rule adds value mostly by keeping exposure to correctly retained firms and avoiding broad over-exclusion. True-positive gains are small because bankrupt/permanent-loss positives are rare, but fixed-FPR rules limit the false-positive cost.

FP and FN are opportunity-cost terms; TP and TN are gains versus the matched benchmark decomposition. Net matches the selected OOS strategy alpha before cost overlays. *Remark:* Source: model-specific

error\_cost.decomposition.by\_crsp.universe.csv; matched to best\_by\_track.index.cost.csv by track, universe, model, threshold, permanent rule, strategy id, and OOS period.



# Transaction-Cost Robustness

Winner ranking is stable from 0 to 20 bps; costs apply to gross buy+sell turnover

| Track | Universe | 0 bps winner    | 20 bps winner   | Alpha 0 | Alpha 20 |
|-------|----------|-----------------|-----------------|---------|----------|
| Temp. | Total    | fund            | fund            | +0.43pp | +0.42pp  |
| Temp. | Large    | fund            | fund            | +0.15pp | +0.15pp  |
| Temp. | Mid      | fund            | fund            | +0.51pp | +0.51pp  |
| Temp. | Small    | raw_plus_latent | raw_plus_latent | +0.64pp | +0.62pp  |
| Perm. | Total    | raw_plus_latent | raw_plus_latent | +0.27pp | +0.27pp  |
| Perm. | Large    | raw_plus_latent | raw_plus_latent | +0.23pp | +0.23pp  |
| Perm. | Mid      | fund            | fund            | +0.74pp | +0.74pp  |
| Perm. | Small    | latent_raw      | latent_raw      | +0.32pp | +0.32pp  |

Transaction costs do not change the top winner in any of the eight track-index comparisons. The cost drag is economically visible in Mid and Small Cap, but it is not large enough at 20 bps to overturn the ranking. *Remark:*

Alpha is net annualized geometric return minus the corresponding benchmark annualized return. Source: `transaction.cost.robustness.summary.csv`.

# Turnover Effect

Best strategies keep their ranking at 10 bps; turnover is gross buy+sell traded weight

| Temporary CSI |                                    |               |                |                 |               |
|---------------|------------------------------------|---------------|----------------|-----------------|---------------|
| Universe      | Best strategy                      | Ann. turnover | Geo ret. 0 bps | Geo ret. 10 bps | Delta from TC |
| Total         | AG Base Dataset (Youden 3yr)       | 15.3%         | 13.72%         | 13.71%          | -0.01pp       |
| Large         | AG Base Dataset (Youden 3yr)       | 18.8%         | 14.43%         | 14.42%          | -0.01pp       |
| Mid           | AG Base Dataset (FPR5 5yr)         | 97.7%         | 10.16%         | 10.05%          | -0.11pp       |
| Small         | AG Exp. Dataset + VAE (Youden 3yr) | 77.2%         | 8.29%          | 8.20%           | -0.09pp       |

| Permanent CSI |                                      |               |                |                 |               |
|---------------|--------------------------------------|---------------|----------------|-----------------|---------------|
| Universe      | Best strategy                        | Ann. turnover | Geo ret. 0 bps | Geo ret. 10 bps | Delta from TC |
| Total         | AG Exp. Dataset + VAE (FPR5 perm.)   | 12.2%         | 13.56%         | 13.55%          | -0.01pp       |
| Large         | AG Exp. Dataset + VAE (FPR5 perm.)   | 17.2%         | 14.51%         | 14.49%          | -0.01pp       |
| Mid           | AG Base Dataset (FPR5 perm.)         | 97.0%         | 10.39%         | 10.28%          | -0.12pp       |
| Small         | AG Latent Dataset (VAE) (FPR3 perm.) | 72.4%         | 7.98%          | 7.90%           | -0.08pp       |

Mid and Small Cap exclusions create the largest annualized gross turnover, measured as buy plus sell traded weight, so cost drag is visibly larger there. The 0–20 bps sweep did not change any top winner ranking. *Remark:*

Source: turnover.summary.csv, winner.turnover.summary.20bps.csv, and best.by.track.index.cost.csv.

# Threshold Families and Turnover

Threshold-family result at 20 bps

| Track | Lead family | Mean alpha    |
|-------|-------------|---------------|
| Temp. | Youden      | +0.35pp       |
| Perm. | FPR3/FPR5   | +0.35/+0.33pp |

Temporary CSI benefits from calibrated finite lockouts; 3–5yr exclusions outperform 1yr exclusions. Permanent CSI needs stricter fixed-FPR rules; Youden is negative on average under absorbing exclusion.

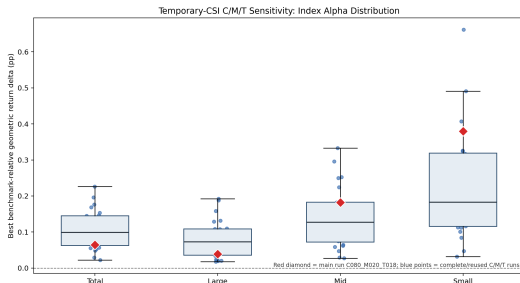
*Remark:* Sources: `threshold_family_summary_20bps.csv` and `winner_turnover_summary_20bps.csv`.

Annualized gross turnover for 20 bps winners

| Universe | Temp. | Perm. |
|----------|-------|-------|
| Total    | 15.3% | 12.2% |
| Large    | 18.8% | 17.2% |
| Mid      | 97.7% | 97.0% |
| Small    | 77.2% | 72.4% |

Turnover matters most in Mid and Small Cap. At 20 bps, the annual return drag is still modest: roughly 1.1pp in Mid Cap and below 0.9pp in Small Cap winners.

# Sensitivity: Main Run Versus C/M/T Grid



## Key results: best alpha distribution

| Univ. | Runs | Main | Median | Range     |
|-------|------|------|--------|-----------|
| Total | 24   | 0.06 | 0.10   | 0.02–0.23 |
| Large | 24   | 0.04 | 0.07   | 0.02–0.19 |
| Mid   | 24   | 0.18 | 0.13   | 0.03–0.33 |
| Small | 24   | 0.38 | 0.18   | 0.03–0.66 |

Values are benchmark-relative geometric return deltas in percentage points.

- The accepted main run sits inside the completed-run distribution.
- Main-run alpha is below the median for Total/Large and above the median for Mid/Small.
- Sensitivity changes magnitude more than direction: completed runs remain positive across all four universes.
- Three blocked partial C/M/T configs remain excluded from completed-run conclusions.

Remark: Sources: chart1.sensitivity\_stability\_distribution.png; universe\_stability\_summary.csv; sensitivity\_index\_stability\_table.csv; AE-SENS-CHART-001R interpretation.

# Remaining Robustness Work

## What is complete vs. still future work

Temporary-CSI C/M/T sensitivity is complete locally and supports the presentation robustness narrative. The remaining items below are extensions or validation work, not completed results.

### **Permanent-CSI sensitivity grid**

Repeat the C/M/T-style robustness logic for the permanent response track before claiming permanent-label sensitivity.

### **Probability-weighted strategy**

Active CSI-probability weighting is not yet implemented; current results use thresholded exclusions.

### **Benchmark extensions**

Minimum-volatility and quality/risk-scaling benchmarks remain future comparator work.

### **Residual terminal-failure diagnostic**

Recovered bankrupt/insolvent firm diagnostics remain a follow-up, not a completed robustness result.

*Remark:* The final index suite and the temporary sensitivity grid answer different questions and are kept separate in the presentation.

# Appendix A1: Delisting and Bankruptcy Detection

| Indicator group                     | CRSP firms | Detected | Rate   | Positive rows followed |
|-------------------------------------|------------|----------|--------|------------------------|
| Bankruptcy range 572-574            | 629        | 545      | 86.65% | 2,757                  |
| Bankruptcy / declared insolvent 574 | 548        | 513      | 93.61% | 2,502                  |
| Adverse 400-490, 550-585            | 5,654      | 3,676    | 65.02% | 7,000                  |

## Why the override is additive

The CRSP signal is not an independent default label. It only turns a CSI-like trigger into a positive `dynamic_csi` event when the firm fails before the 18-month confirmation clock can fully resolve.

# Appendix A2: Revised Annual Prevalence

## Current CRSP 572–574 overlap

| Annual label metric                     |       | Metric                          |        |
|-----------------------------------------|-------|---------------------------------|--------|
|                                         | Count |                                 | Count  |
| Temporary positive label cells          | 8,517 | CRSP 572–574 firms in sample    | 629    |
| Temporary unresolved cells ( $y = NA$ ) | 8,674 | Detected by revised temporary   | 545    |
| Permanent positive label cells          | 6,258 | CSI                             |        |
| Permanent unresolved cells ( $y = NA$ ) | 834   | Missed by revised temporary CSI | 84     |
|                                         |       | Detection rate                  | 86.65% |

The remaining misses are not treated as current pipeline errors. Under the additive rule, CRSP bankruptcy codes only validate an already triggered CSI event within the timing window; they are not standalone default labels.

## Methodological status

The current overlap audit supports the 545/84/86.65% story. A full reason breakdown for all 84 CRSP 572–574 misses would require a separate diagnostic; the current slide treats them as exclusions under the retained trigger-and-window definition.

# Appendix A3: Cleaned Label Counts

| CSI type  | Split | Obs. rows | $y = 0$ | $y = 1$      | $y = NA$ | Labelled | Prev. |
|-----------|-------|-----------|---------|--------------|----------|----------|-------|
| Temporary | Full  | 188,460   | 171,269 | <b>8,517</b> | 8,674    | 179,786  | 4.74% |
| Temporary | Train | 143,173   | 136,630 | 6,543        | 0        | 143,173  | 4.57% |
| Temporary | Test  | 18,111    | 17,307  | 804          | 0        | 18,111   | 4.44% |
| Temporary | OOS   | 27,176    | 17,332  | 1,170        | 8,674    | 18,502   | 6.32% |
| Permanent | Full  | 188,460   | 181,368 | 6,258        | 834      | 187,626  | 3.34% |
| Permanent | Train | 143,173   | 137,794 | 5,379        | 0        | 143,173  | 3.76% |
| Permanent | Test  | 18,111    | 17,511  | 542          | 58       | 18,053   | 3.00% |
| Permanent | OOS   | 27,176    | 26,063  | 337          | 776      | 26,400   | 1.28% |

## Read

Both targets remain rare after the observable-scaffold cleanup.  $y = NA$  rows stay in canonical artifacts but are excluded from supervised training and label-based evaluation.

Source: Labelled rows =  $y = 0 + y = 1$ ; prevalence =  $y = 1$ /labelled rows.



# Appendix A4: Descriptive Statistics – Market Cap

Market cap, USD millions

| CSI type  | Split | y | Mean    | Median | P25  | P75     |
|-----------|-------|---|---------|--------|------|---------|
| Temporary | CV    | 0 | 4,117.4 | 408.1  | 87.3 | 1,924.5 |
| Temporary | CV    | 1 | 613.4   | 85.1   | 30.0 | 255.5   |
| Temporary | Full  | 0 | 4,864.6 | 350.9  | 67.2 | 1,903.7 |
| Temporary | Full  | 1 | 574.6   | 68.7   | 23.3 | 228.0   |
| Permanent | CV    | 0 | 3,452.1 | 354.5  | 85.2 | 1,551.6 |
| Permanent | CV    | 1 | 447.3   | 79.6   | 30.3 | 221.6   |
| Permanent | Full  | 0 | 3,558.3 | 249.3  | 57.7 | 1,232.6 |
| Permanent | Full  | 1 | 496.5   | 66.2   | 22.7 | 219.0   |

## Read

CSI-positive firms are much smaller across both definitions. In the full sample, median market cap is 68.7 for temporary positives versus 350.9 for non-positives, and 66.2 for permanent positives versus 249.3 for non-positives.

Source: Market cap is USD millions.

## Appendix A5: Descriptive Statistics – Other Medians

| CSI type  | Split | y | Ann.<br>ret. | Assets | Sales | ROA    | M/B  |
|-----------|-------|---|--------------|--------|-------|--------|------|
| Temporary | CV    | 0 | 6.1%         | 640.9  | 326.7 | 1.7%   | 1.82 |
| Temporary | CV    | 1 | -24.9%       | 86.1   | 50.2  | -14.2% | 1.51 |
| Temporary | Full  | 0 | 4.9%         | 565.3  | 295.2 | 1.6%   | 1.81 |
| Temporary | Full  | 1 | -35.0%       | 77.5   | 46.8  | -18.8% | 1.59 |
| Permanent | CV    | 0 | 6.0%         | 567.8  | 270.2 | 1.7%   | 1.82 |
| Permanent | CV    | 1 | -28.3%       | 78.8   | 44.4  | -15.8% | 1.46 |
| Permanent | Full  | 0 | 4.2%         | 395.3  | 191.9 | 1.5%   | 1.80 |
| Permanent | Full  | 1 | -37.5%       | 64.6   | 35.1  | -19.3% | 1.52 |

### Read

The positive class has sharply lower realized returns, smaller operating scale, and negative profitability. The contrast strengthens in the full sample for both temporary and permanent CSI.

Source: Assets and sales are USD millions.

# Appendix A6: CV Folds, Leakage Controls, and Training Metric

## Expanding-window CV

| Fold | Training years         | Validation years                             |
|------|------------------------|----------------------------------------------|
| 1    | Initial training block | No validation; used to seed expanding window |
| 2    | Up to 1998             | 1999-2004                                    |
| 3    | Up to 2004             | 2005-2009                                    |
| 4    | Up to 2009             | 2010-2015                                    |

## Training / selection metric

- **AutoGluon:** yes, model ranking uses AP via `EVAL_METRIC = "average_precision"` passed to `TabularPredictor(..., eval_metric=EVAL_METRIC)` in `09C_AutoGluon.py`.
- **XGBoost:** binary-logistic training uses `eval_metric="aucpr"` in `09_Train.R`; reported AP is computed from predicted probabilities.
- Split before preprocessing.
- Train-fitted winsorization, imputation, and PIT only.
- CV AP/AUC/R@FPR metrics are averaged per fold; OOS is reserved for index construction, not model selection.

# Appendix A7: Feature Family Inventory

| Family            | Examples / purpose                                                                                                                 |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Fundamentals      | Balance sheet, income statement, profitability, leverage, liquidity, accounting dynamics.                                          |
| Market and price  | Returns, drawdowns, volatility, momentum, market capitalization, price-path information.                                           |
| Macro             | FRED macro series and macro interaction terms.                                                                                     |
| Latent            | VAE latent factors from fundamentals or raw features plus reconstruction error.                                                    |
| Metadata excluded | Labels, event dates, confirmation dates, censoring flags, and diagnostics are retained for audit but excluded from model features. |

# Appendix A8: VAE Architecture Details

| Component        | Setting                                                                                       |
|------------------|-----------------------------------------------------------------------------------------------|
| Encoder          | $n\_features \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow z\_mean, z\_log\_var$ |
| Latent dimension | 24                                                                                            |
| Decoder          | $z \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow n\_features$                    |
| Loss             | Reconstruction + $\beta$ KL + $\gamma$ supervised CSI classification loss                     |
| Parameters       | $\beta = 1.0, \gamma = 0.1$ , early stopping patience 15                                      |
| Status           | Weakly supervised on training rows; test and OOS labels are not used for VAE fitting.         |

# Appendix A9: Temporary CSI CV/Test/OOS Model Metrics

Full model-suite threshold metrics    Train/CV, test, and OOS; AP/AUC/FPR recall.

| Split | Model                   | AP            | AUC           | R@FPR1        | R@FPR3        | R@FPR5        | Brier         |
|-------|-------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| CV    | AG Expanded Dataset     | 0.2046        | 0.8635        | 0.0917        | 0.2344        | 0.3415        | 0.0371        |
| CV    | AG Base Dataset         | 0.1923        | 0.8446        | 0.0830        | 0.2206        | 0.3229        | 0.0374        |
| CV    | AG Latent Dataset (VAE) | 0.1659        | 0.8454        | 0.0670        | 0.1706        | 0.2735        | 0.0380        |
| CV    | AG Exp. Dataset + VAE   | <b>0.2114</b> | <b>0.8667</b> | <b>0.0981</b> | <b>0.2478</b> | <b>0.3581</b> | <b>0.0368</b> |
| Test  | AG Expanded Dataset     | <b>0.1985</b> | <b>0.8766</b> | <b>0.0821</b> | <b>0.2002</b> | <b>0.2998</b> | <b>0.0389</b> |
| Test  | AG Base Dataset         | 0.1656        | 0.8538        | 0.0585        | 0.1517        | 0.2413        | 0.0392        |
| Test  | AG Latent Dataset (VAE) | 0.1501        | 0.8438        | 0.0410        | 0.1294        | 0.2264        | 0.0394        |
| Test  | AG Exp. Dataset + VAE   | 0.1864        | 0.8741        | 0.0634        | 0.1779        | 0.2749        | 0.0394        |
| OOS   | AG Expanded Dataset     | 0.3084        | <b>0.8961</b> | 0.0915        | 0.2547        | 0.4000        | 0.0498        |
| OOS   | AG Base Dataset         | 0.2526        | 0.8686        | 0.0598        | 0.1795        | 0.3017        | 0.0526        |
| OOS   | AG Latent Dataset (VAE) | 0.2813        | 0.8626        | 0.1009        | 0.2274        | 0.3556        | 0.0518        |
| OOS   | AG Exp. Dataset + VAE   | <b>0.3152</b> | 0.8950        | <b>0.1051</b> | <b>0.2632</b> | <b>0.4128</b> | <b>0.0494</b> |

Remark: Source: 03\_Data\_Output/6\_ModelSuite/comparison/AE-MODEL-SUITE-007\_model\_suite\_metrics\_long.csv; raw comparator from raw/derived\_metrics/raw\_complete\_threshold\_metrics.csv.

# Appendix A10: Permanent CSI CV/Test/OOS Model Metrics

Full model-suite threshold metrics    Train/CV, test, and OOS; AP/AUC/FPR recall.

| Split | Model                   | AP            | AUC           | R@FPR1        | R@FPR3        | R@FPR5        | Brier         |
|-------|-------------------------|---------------|---------------|---------------|---------------|---------------|---------------|
| CV    | AG Expanded Dataset     | 0.1854        | <b>0.8757</b> | 0.1098        | <b>0.2607</b> | <b>0.3697</b> | 0.0297        |
| CV    | AG Base Dataset         | 0.1785        | 0.8677        | 0.1033        | 0.2379        | 0.3469        | 0.0299        |
| CV    | AG Latent Dataset (VAE) | 0.1426        | 0.8460        | 0.0732        | 0.1960        | 0.2908        | 0.0308        |
| CV    | AG Exp. Dataset + VAE   | <b>0.1883</b> | 0.8719        | <b>0.1196</b> | 0.2578        | 0.3680        | <b>0.0297</b> |
| Test  | AG Expanded Dataset     | <b>0.1416</b> | 0.8810        | <b>0.0664</b> | 0.1808        | 0.3007        | 0.0284        |
| Test  | AG Base Dataset         | 0.1289        | 0.8627        | 0.0627        | 0.1697        | 0.2620        | 0.0293        |
| Test  | AG Latent Dataset (VAE) | 0.1115        | 0.8496        | 0.0424        | 0.1513        | 0.2546        | <b>0.0280</b> |
| Test  | AG Exp. Dataset + VAE   | 0.1415        | <b>0.8838</b> | 0.0609        | <b>0.2011</b> | <b>0.3155</b> | 0.0283        |
| OOS   | AG Expanded Dataset     | 0.0323        | 0.8081        | 0.0000        | 0.0119        | 0.0623        | 0.0211        |
| OOS   | AG Base Dataset         | 0.0281        | 0.7798        | 0.0000        | 0.0030        | 0.0297        | 0.0232        |
| OOS   | AG Latent Dataset (VAE) | <b>0.0482</b> | <b>0.8337</b> | <b>0.0504</b> | <b>0.1484</b> | <b>0.2226</b> | <b>0.0166</b> |
| OOS   | AG Exp. Dataset + VAE   | 0.0311        | 0.8032        | 0.0000        | 0.0208        | 0.0653        | 0.0207        |

Remark: Source: 03\_Data\_Output/6\_ModelSuite/comparison/AE-MODEL-SUITE-007\_model\_suite\_metrics\_long.csv; raw comparator from raw/derived\_metrics/raw\_complete\_threshold\_metrics.csv.

# Appendix A11: Model Family Winners

## AutoGluon selection evidence

Every final feature-set/track run selected a `WeightedEnsemble_L2` as the top leaderboard model. The useful base signal is mostly tree-based; neural nets were considered but did not dominate.

| Track     | Feature set             | Best family      | Val. score    | Fit sec. |
|-----------|-------------------------|------------------|---------------|----------|
| Temporary | AG Expanded Dataset     | WeightedEnsemble | 0.2307        | 280      |
| Temporary | AG Base Dataset         | WeightedEnsemble | 0.2203        | 83       |
| Temporary | AG Latent Dataset (VAE) | WeightedEnsemble | 0.1953        | 146      |
| Temporary | AG Exp. Dataset + VAE   | WeightedEnsemble | <b>0.2314</b> | 203      |
| Permanent | AG Expanded Dataset     | WeightedEnsemble | 0.2010        | 126      |
| Permanent | AG Base Dataset         | WeightedEnsemble | 0.1929        | 229      |
| Permanent | AG Latent Dataset (VAE) | WeightedEnsemble | 0.1505        | 191      |
| Permanent | AG Exp. Dataset + VAE   | WeightedEnsemble | <b>0.2041</b> | 291      |

Remark: Source: AE-MODEL-SUITE-007\_model\_family\_winners.csv.



# Appendix A12: Model Metric Caveats and OOS Robustness

## Comparability

The raw comparator is the revised-dataset optional-library AutoGluon rerun from AE-VALIDATE. Complete local threshold metrics were later materialized under `03_Data_Output/6_ModelSuite/raw/derived_metrics`, so the presentation uses AP, AUC, Brier, and R@FPR1/3/5 consistently across feature sets.

- AP and fixed-FPR recall are emphasized because CSI positives are rare.
- AUC remains useful as a ranking diagnostic but can mask low-prevalence false-positive economics.
- AG Exp. Dataset + VAE is the main enhanced feature-set challenger for temporary CSI and permanent test/CV.
- AG Latent Dataset (VAE) is retained for permanent OOS robustness, not as a universal replacement.
- Economic superiority is not claimed from model metrics alone; index-construction slides test benchmark-relative performance.

# Appendix A13: Final Index Grid Contract

## Completed grid

- Models: raw (AG Expanded Dataset), fund (AG Base Dataset), latent\_raw (AG Latent Dataset), raw\_plus\_latent (AG Exp. Dataset + VAE).
- Tracks: temporary CSI and permanent CSI.
- Universes: Total Market, Large Cap, Mid Cap, Small Cap.
- Thresholds: Youden, FPR1, FPR3, FPR5.

## Portfolio rules

- Temporary CSI: 1-, 2-, 3-, and 5-year exclusion lockouts.
- Permanent CSI: absorbing permanent removal after first active signal.
- Costs: 0, 5, 10, and 20 bps on gross turnover.
- Turnover output is populated for every model-track-universe cell.

## Presentation interpretation

The index section uses OOS economic performance. Model metrics identify useful prediction signals; index construction tests whether those signals improve benchmark-relative returns after thresholding, exclusions, turnover, and transaction costs.

*Remark:* Sources: `comparison/full_grid.manifest.csv`; `comparison/best.by_track_index.cost.csv`; `comparison/turnover.summary.csv`.

# Appendix A14: 20 bps Winners With Benchmarks

Best strategy by track and universe at 20 bps

| Track | Universe | Winner          | Rule         | Net ret. | Bench. | Alpha   |
|-------|----------|-----------------|--------------|----------|--------|---------|
| Temp. | Total    | fund            | Youden / 3yr | 13.70%   | 13.28% | +0.42pp |
| Temp. | Large    | fund            | Youden / 3yr | 14.40%   | 14.25% | +0.15pp |
| Temp. | Mid      | fund            | FPR5 / 5yr   | 9.93%    | 9.42%  | +0.51pp |
| Temp. | Small    | raw_plus_latent | Youden / 3yr | 8.11%    | 7.49%  | +0.62pp |
| Perm. | Total    | raw_plus_latent | FPR5 / perm. | 13.54%   | 13.28% | +0.27pp |
| Perm. | Large    | raw_plus_latent | FPR5 / perm. | 14.48%   | 14.25% | +0.23pp |
| Perm. | Mid      | fund            | FPR5 / perm. | 10.16%   | 9.42%  | +0.74pp |
| Perm. | Small    | latent_raw      | FPR3 / perm. | 7.82%    | 7.49%  | +0.32pp |

VAE/non-raw variants beat raw in 5 of 8 track-index cases at 20 bps. The market-weighted benchmark is shown as the return reference and does not receive a strategy cost overlay. *Remark:* Source:

`presentation_headline_tables.md`; final claim from AE-INDEX-SUITE-009\_Closeout\_Report.md.

# Appendix A15: Error Decomposition and Index Source Paths

**Base folders:** 03\_Data\_Output/7\_IndexConstructionValidation/nonraw\_index\_suite/

- **Headline winners:** final\_tables/headline\_winners\_20bps.csv
- **Presentation tables:** final\_tables/presentation\_headline\_tables.md
- **Transaction costs:** comparison/transaction\_cost\_impact.csv;  
final\_tables/transaction\_cost\_robustness\_summary.csv
- **Best-by-cost grid:** comparison/best\_by\_track\_index\_cost.csv
- **Threshold-family grid:** comparison/best\_by\_track\_index\_threshold\_family.csv
- **Strategy performance fields:** model-specific index\_performance\_gross\_and\_net\_by\_tc.csv
- **Threshold families:** final\_tables/threshold\_family\_summary\_20bps.csv
- **Turnover:** comparison/turnover\_summary.csv; final\_tables/winner\_turnover\_summary\_20bps.csv
- **Error-cost decomposition:** selected fund, raw\_plus\_latent, and latent\_raw  
error\_cost\_decomposition\_by\_crsp\_universe.csv files mapped exactly in SLIDE\_DATA\_SOURCES.md.

## Source discipline

All claims in the index-result slides are sourced from the completed AE-INDEX-SUITE grid. Sensitivity-analysis results are intentionally not finalized in this section.

# Appendix A16: Transaction-Cost Robustness

Top winner did not change between 0 and 20 bps

| Track | Universe | Winner 0 bps    | Winner 20 bps   | Alpha 0 | Alpha 20 |
|-------|----------|-----------------|-----------------|---------|----------|
| Temp. | Total    | fund            | fund            | +0.43pp | +0.42pp  |
| Temp. | Large    | fund            | fund            | +0.15pp | +0.15pp  |
| Temp. | Mid      | fund            | fund            | +0.51pp | +0.51pp  |
| Temp. | Small    | raw_plus_latent | raw_plus_latent | +0.64pp | +0.62pp  |
| Perm. | Total    | raw_plus_latent | raw_plus_latent | +0.27pp | +0.27pp  |
| Perm. | Large    | raw_plus_latent | raw_plus_latent | +0.23pp | +0.23pp  |
| Perm. | Mid      | fund            | fund            | +0.74pp | +0.74pp  |
| Perm. | Small    | latent_raw      | latent_raw      | +0.32pp | +0.32pp  |

Costs are applied at 0, 5, 10, and 20 bps. The 20 bps case is conservative enough for presentation, but not high enough to reverse the headline rankings. *Remark:* Source: transaction\_cost\_robustness\_summary.csv.

# Appendix A17: Threshold Family and Turnover Summary

Mean best alpha by threshold family

| Track | Threshold | Mean alpha |
|-------|-----------|------------|
| Temp. | Youden    | +0.35pp    |
| Temp. | FPR3      | +0.19pp    |
| Temp. | FPR5      | +0.17pp    |
| Temp. | FPR1      | +0.12pp    |
| Perm. | FPR3      | +0.35pp    |
| Perm. | FPR5      | +0.33pp    |
| Perm. | FPR1      | +0.14pp    |
| Perm. | Youden    | -0.20pp    |

Mean annualized turnover for winners

| Universe | Temp. | Perm. |
|----------|-------|-------|
| Total    | 15.3% | 12.2% |
| Large    | 18.8% | 17.2% |
| Mid      | 97.7% | 97.0% |
| Small    | 77.2% | 72.4% |

Temporary CSI is strongest with Youden and longer finite lockouts; permanent CSI is strongest with FPR3/FPR5. Turnover is concentrated in Mid and Small Cap, where exclusions shift more of the investable universe. *Remark:* Sources: `threshold_family_summary_20bps.csv`; `winner_turnover_summary_20bps.csv`.

# Appendix A18: Temporary CSI Sensitivity Detail

## Selected configurations from the 27-run temporary-CSI sensitivity grid

| Role       | Run ID         | Pos.   | Prev.  | OOS<br>AP | OOS<br>AUC | 11C TM<br>alpha | Status   |
|------------|----------------|--------|--------|-----------|------------|-----------------|----------|
| Composite  | C090_M000.T012 | 12,761 | 7.02%  | 0.557     | 0.920      | +0.169pp        | complete |
| AP winner  | C060_M000.T012 | 31,596 | 17.49% | 0.566     | 0.864      | +0.052pp        | reused   |
| 11C winner | C090_M020.T018 | 6,035  | 3.34%  | 0.274     | 0.911      | +0.226pp        | complete |
| Baseline   | C080_M020.T018 | 8,517  | 4.74%  | 0.308     | 0.896      | +0.065pp        | reused   |

## Blocked partial cases

The grid represents 27 C/M/T configurations: 24 are complete or reused, and three are blocked.partial: C080\_M000.T012, C080\_M000.T018, and C060\_M020.T028. Blocked cases are represented in manifests but excluded from completed-run conclusions.

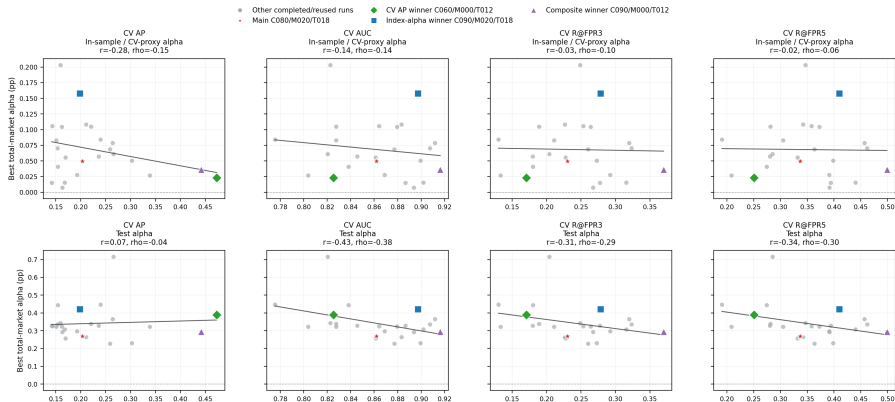
## Source interpretation

C/M/T sensitivity tests label-threshold robustness for temporary CSI. It supports the robustness narrative but does not alter the final accepted label definition used by AE-INDEX-SUITE. In the accepted-label temporary index grid, 0/5/10/20 bps transaction-cost overlays do not change the winner set: Base Dataset wins Total/Large/Mid and Exp. Dataset + VAE wins Small. Costs are applied to annualized gross buy+sell turnover.

*Remark:* Sources: local\_sensitivity.cmt\_model.summary.csv; sensitivity.cmt\_index.summary.csv; temporary\_index.transaction\_cost.lines.csv; temporary.blocked.config.disclosure.csv.

# Appendix A19: CV Metrics Versus Index Alpha

CV model metrics versus total-market index alpha across 24 completed temporary-CSI C/M/T runs



ML metrics are CV-only. Alpha uses the best non-benchmark total-market 11C strategy within each run and period. Insample is labelled as CV-proxy because the 11C source period is named insample, not CV.

- CV-only AP/AUC/R@FPR3/R@FPR5 do not map cleanly into total-market alpha; AP is near zero against test alpha.
- Index performance is the realized result after thresholding, timing, universe weights, and returns, not a generic ML metric.
- The 11C source period is named insample; this slide labels it as in-sample/CV-proxy alpha.

Remark: Sources: [slide48\\_cv\\_metrics\\_vs\\_cv\\_test\\_alpha.png](#); [local\\_sensitivity\\_cmt\\_model\\_summary.csv](#); [combined\\_11c\\_performance.csv](#); AE-PRES-QA-FIX-008 evidence.



# Appendix A20: Future Robustness and Extensions

## Not completed results

The items below are explicitly future validation or extensions. They should not be read as completed empirical findings in this deck.

| Item                                    | Purpose                                                                 | Current status          |
|-----------------------------------------|-------------------------------------------------------------------------|-------------------------|
| Permanent-CSI sensitivity grid          | Test label-threshold robustness for the permanent response track.       | Planned / not completed |
| Minimum-volatility benchmark            | Check whether CSI screens add value beyond a low-risk index comparator. | Planned / not completed |
| Quality/risk-scaling benchmark          | Compare exclusions with quality or risk-scaled benchmark construction.  | Planned / not completed |
| Active CSI-probability weighting        | Use CSI probabilities continuously rather than thresholded exclusions.  | Planned / not completed |
| Recovered bankrupt/insolvent diagnostic | Decompose remaining terminal-failure misses and recovery cases.         | Planned / not completed |

*Remark:* Source status is mapped in SLIDE\_DATA\_SOURCES.md; these are future-work notes, not result claims.

# Appendix A21: Slide-to-Source Audit Map

## Audit artifact

The complete slide-to-source map is stored beside this June source deck:

`06_Presentations/02_FinalPresentation/Necessary/FinalPresentation_June/SLIDE_DATA_SOURCES.md`.

- Dataset and label slides map to AE-PANEL evidence and descriptive-statistics CSVs.
- Model slides map to AE-MODEL-SUITE comparison files and threshold metrics.
- Index slides map to AE-INDEX-SUITE final-table and comparison outputs.
- Sensitivity slides map to local AE-SENS manifests and comparison artifacts.
- Conceptual or future-work slides are marked as such rather than linked to empirical result files.

*Remark:* This appendix frame is a navigation pointer; the source map contains per-frame paths and source type.

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